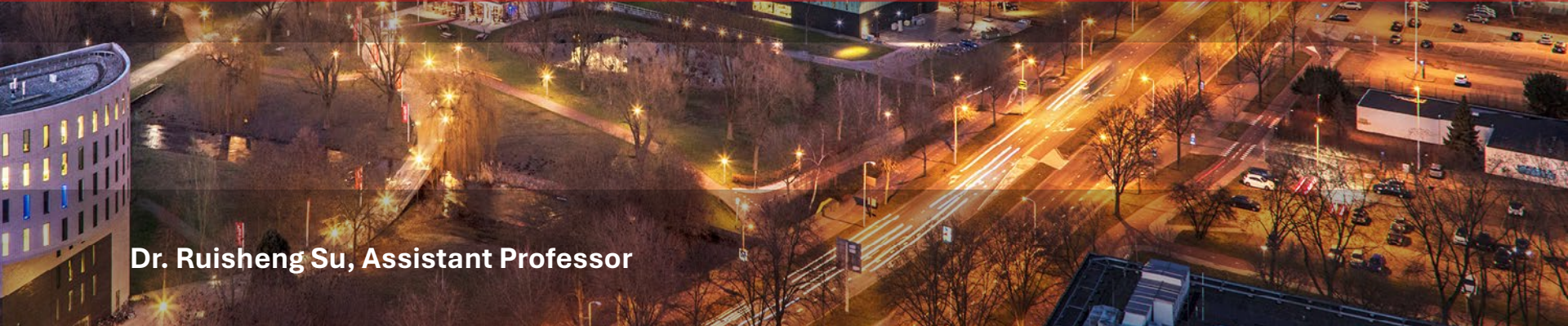




Point-based Image Registration



Dr. Ruisheng Su, Assistant Professor

Department of Biomedical Engineering, Medical Image Analysis group

Overview & course schedule

Modules	Date	Lecturer	Topic
Registration	April 20 (Monday)	C. Scannell & R. Su	Course introduction, geometrical transformations
	April 23 (Thursday)	R. Su	Point-based image registration
	April 30 (Thursday)	R. Su	Intensity-based image registration
Segmentation	May 7 (Thursday)	R. Su	Introduction to image segmentation
	May 11 (Monday)	R. Su	Segmentation in feature space
	May 18 (Monday)	C. Scannell	Segmentation using graph-cuts
	May 21 (Thursday)	C. Scannell	Statistical shape models
Deep learning for MIA	May 28 (Thursday)	C. Scannell	Convolutional neural networks
	June 1 (Monday)	R. Su	Deep learning applications (registration)
	June 4 (Thursday)	A. van Engelen	Industrial guest lecture by (Deep Health)
	June 8 (Monday)	C. Scannell	Deep learning applications (segmentation)
	June 11 (Thursday)	C. Scannell	Unsupervised deep learning for medical image analysis

More details: <https://github.com/tueimage/8BE030-MIA>

Recap

- name possible causes of misalignment in medical images
- name different applications of medical image registration
- classify medical image registration methods using eight different criteria
- apply the basic principles of linear algebra (i.e., matrix-vector, vector-matrix products, transpose, norms, orthogonality, determinant) to image registration tasks
- use the determinant of a transformation matrix T to predict the orientation and magnification of an object transformed with T
- compose and combine rigid and affine transformations in 2D and 3D (and rewrite them using homogeneous coordinates)
- explain the difference between affine and non-linear registrations

Recap

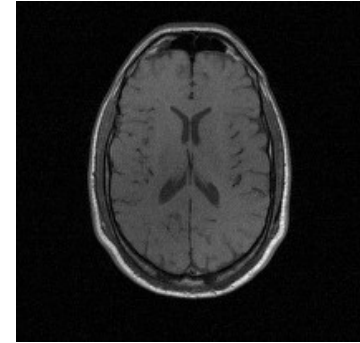
Quiz: Which of the following is not a typical cause of misalignment?

- A. Distortions caused by the imaging system
- B. Patient's voluntary motion during scan
- C. Changes in viewing angle between sessions
- D. Fluctuations in scanner room lighting

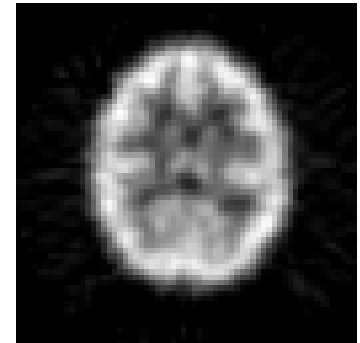
Recap

Quiz: We register the MRI and PET images of the same patient. Which of the following is correct about the category of registration?

- A. Inter-patient
- B. Intra-modal
- C. Inter-modal
- D. Atlas-based



MRI, information about anatomical structures



PET, information about function

Recap

Quiz: An object is transformed using the following transformation matrix: $\begin{bmatrix} 2 & 0 \\ 0 & -0.5 \end{bmatrix}$.
What can we derive about the area and orientation of the transformed object?

- A. Area magnified
- B. Area Shrunk
- C. Rotated
- D. Reflected

Recap

Quiz: Which of the following does **not** necessarily hold for a rotation matrix R ?

A. $RR^{-1} = I$

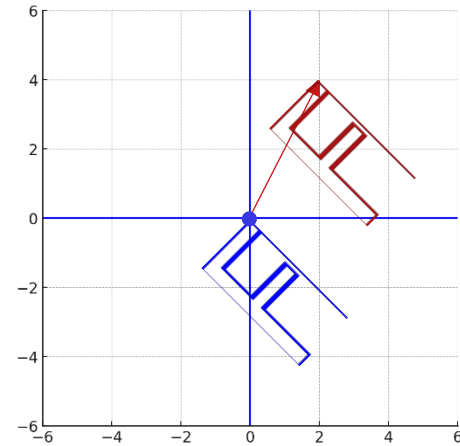
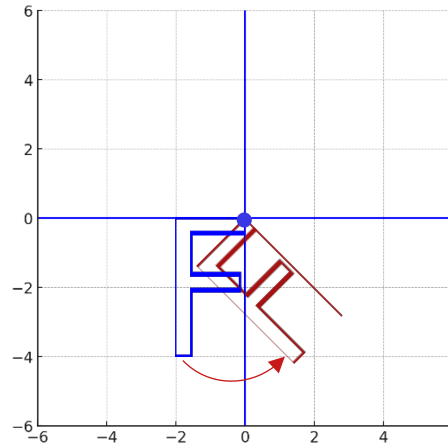
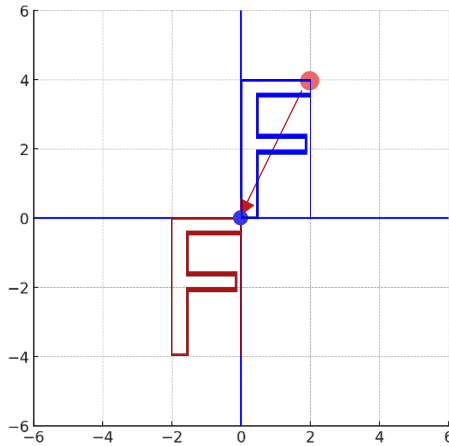
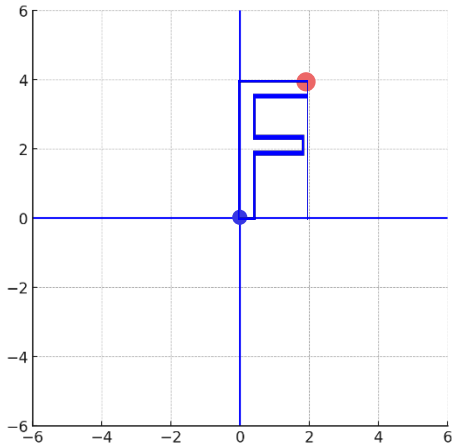
B. $RR^T = I$

C. $R = R^T$

D. $\det(R) = 1$

Recap – Rotation around an arbitrary point

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & x_r \\ 0 & 1 & y_r \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\phi) & -\sin(\phi) & 0 \\ \sin(\phi) & \cos(\phi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -x_r \\ 0 & 1 & -y_r \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



Outline

- Image transformation
- Point-based registration
- Evaluation metrics for point-based registration

Intended learning outcomes – this lecture

- explain the role of inverse mapping and interpolation when transforming an image and compute inverse transformation matrices
- design a general algorithm to register two images based on fiducials (i.e., composing an error function and minimizing this function w.r.t. the transformation T)
- use optimization to find the minimum of this error function
- recall the exact solution for T (matrix notation) when constrained to affine registration
- give at least four reasons why perfect alignment of multiple fiducials is not possible in practice
- describe the algorithm required to find the optimal parameters of T when constrained to rigid registration (orthogonal Procrustes problem)
- explain the principle behind the iterative closest point algorithm
- use the target registration error (TRE) to evaluate image registration

Outline

- Image transformation
- Point-based registration
- Evaluation metrics for point-based registration

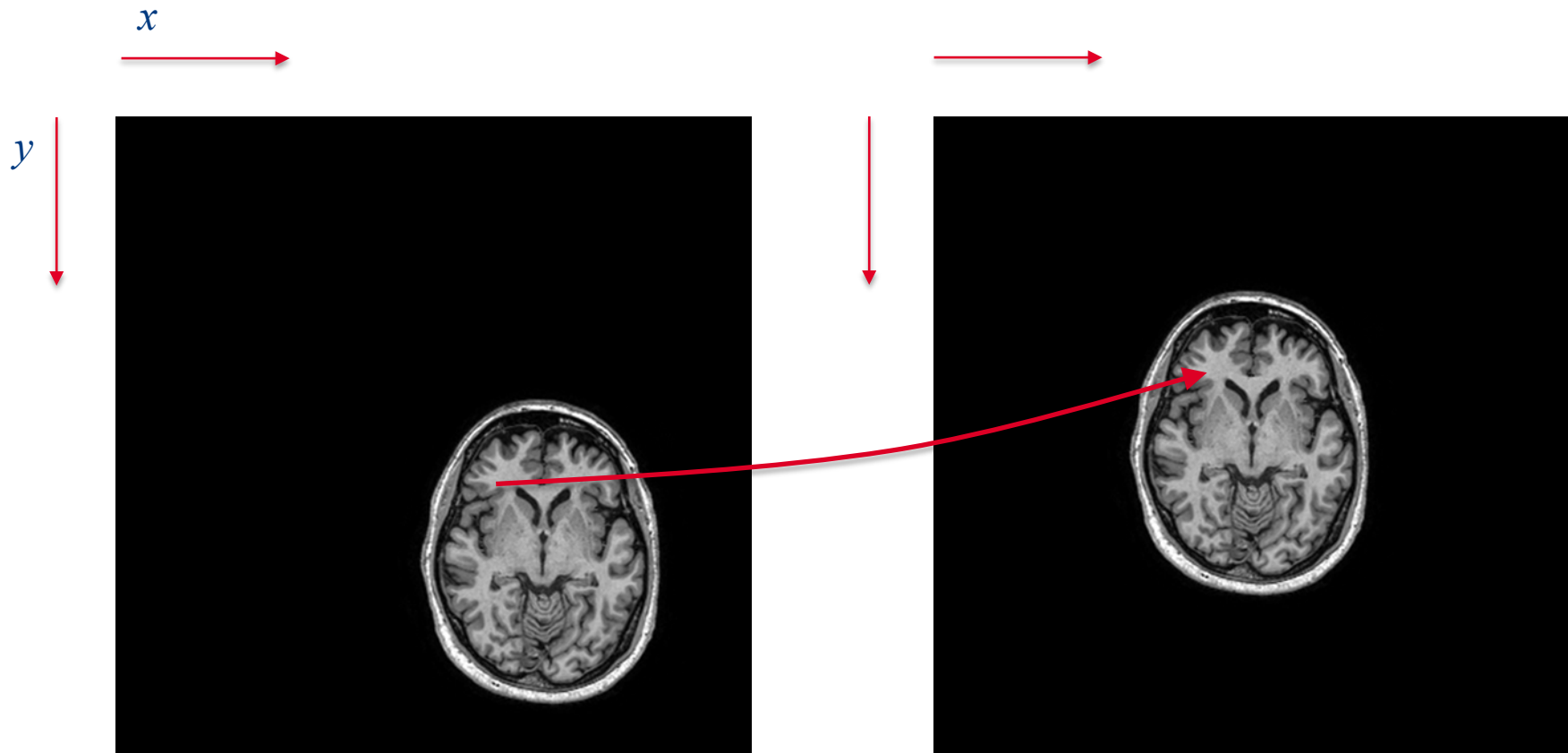
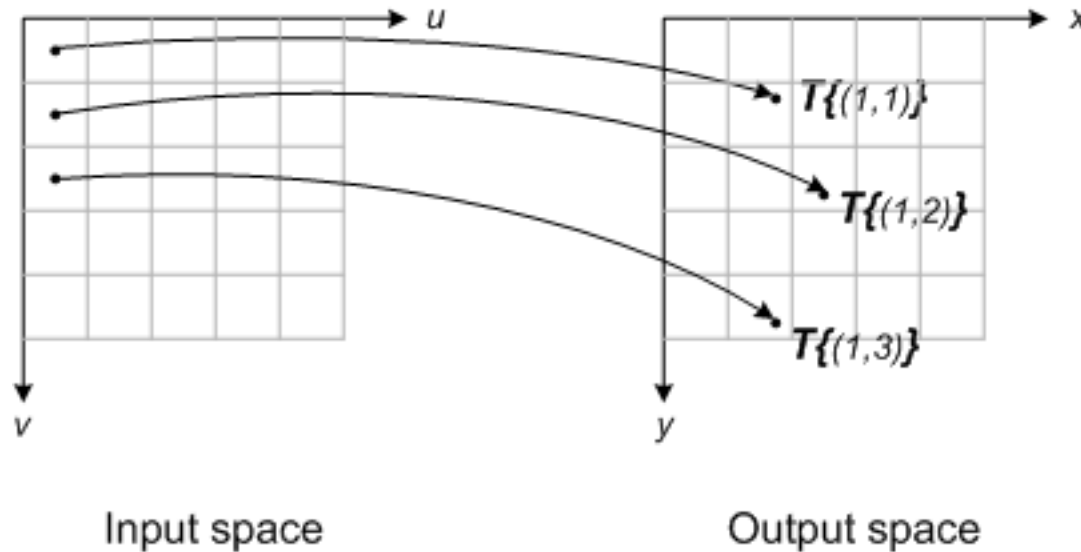


Image transformation

Transforming an image means transforming the spatial coordinates of the pixels.

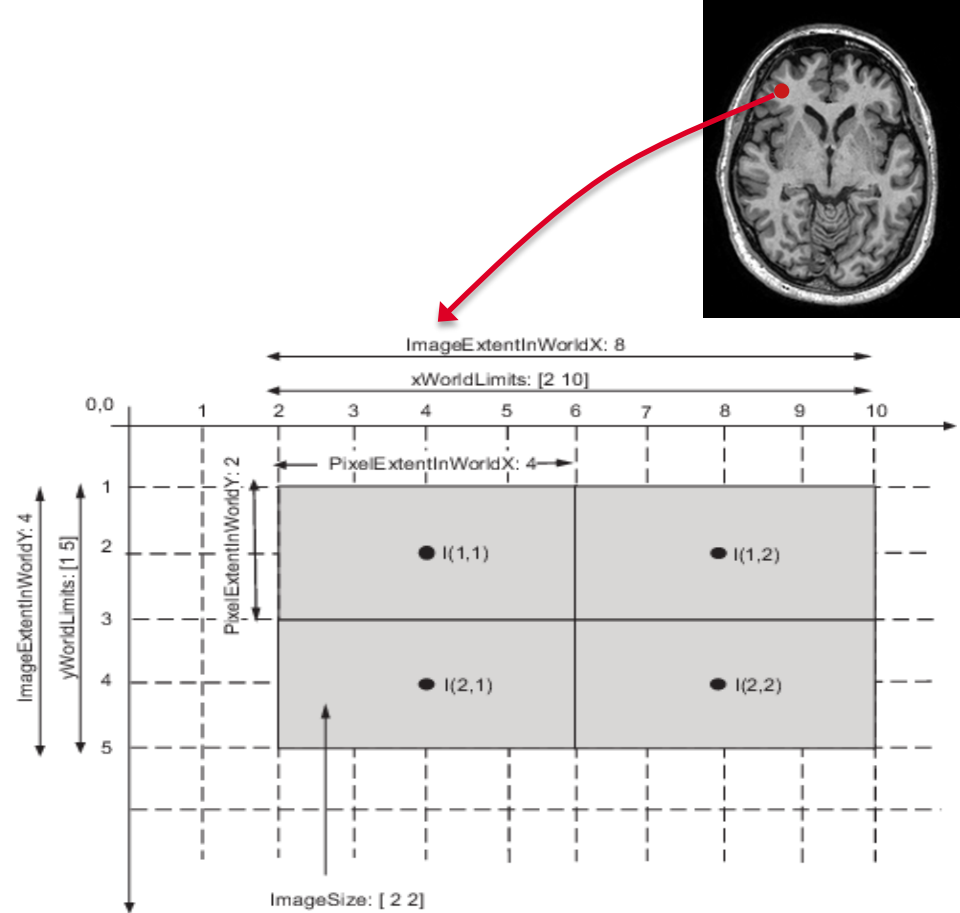


Pixels

Images are stored as arrays where each element corresponds to a pixel intensity value.

In addition to the **intensity**, in medical imaging each pixel is associated with:

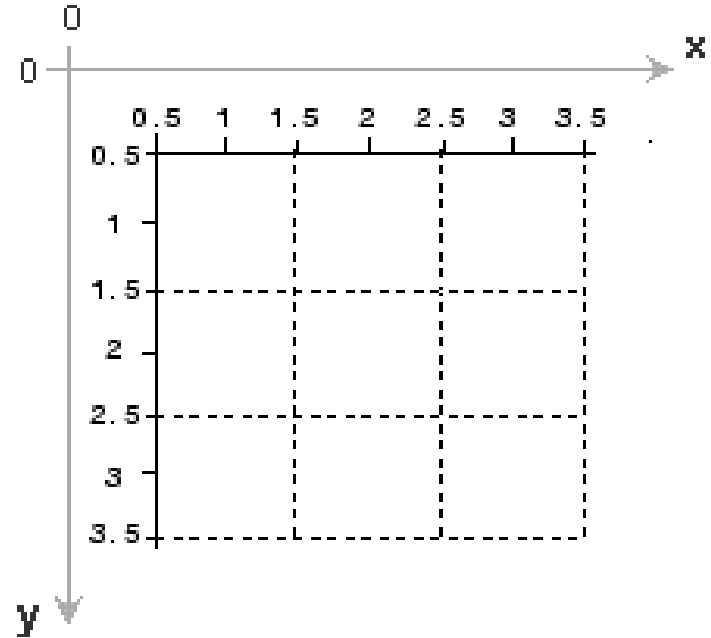
- **Spatial coordinates** – the coordinates in some world coordinate system where the pixel intensity value “appears”.
- **Extent** – the physical size of the pixel.



Assumptions

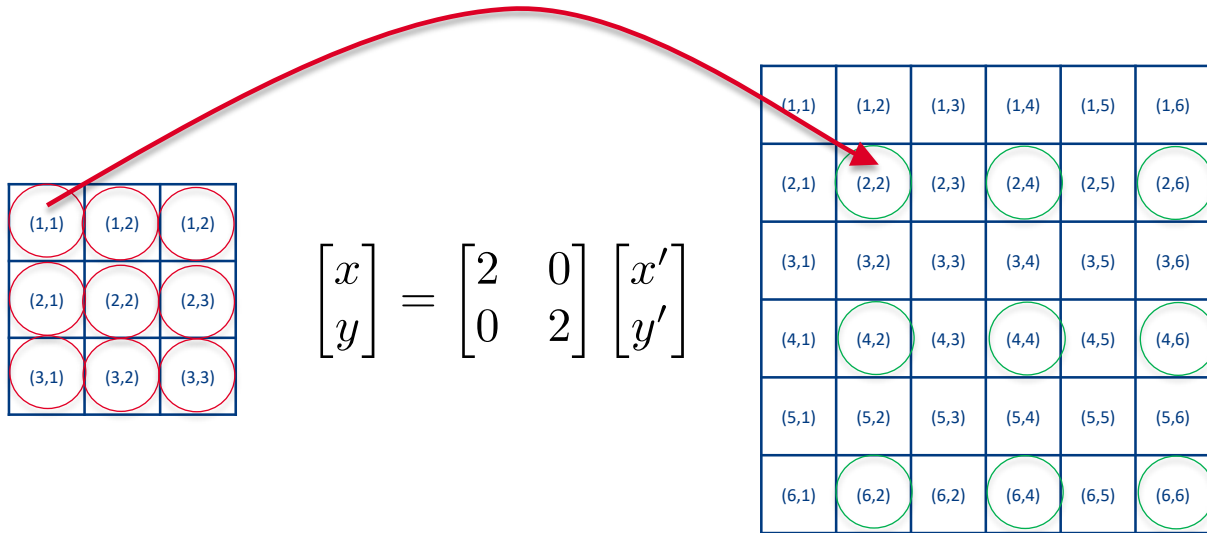
- To simplify the discussion and the implementation in practice, we are going to assume that the pixel indices correspond to the spatial coordinates.
- Furthermore, we are assuming that all images have pixels of the same size and shape (unit size isotropic).

However, note that in practical applications the concepts of physical pixel size and spatial coordinates are very important.



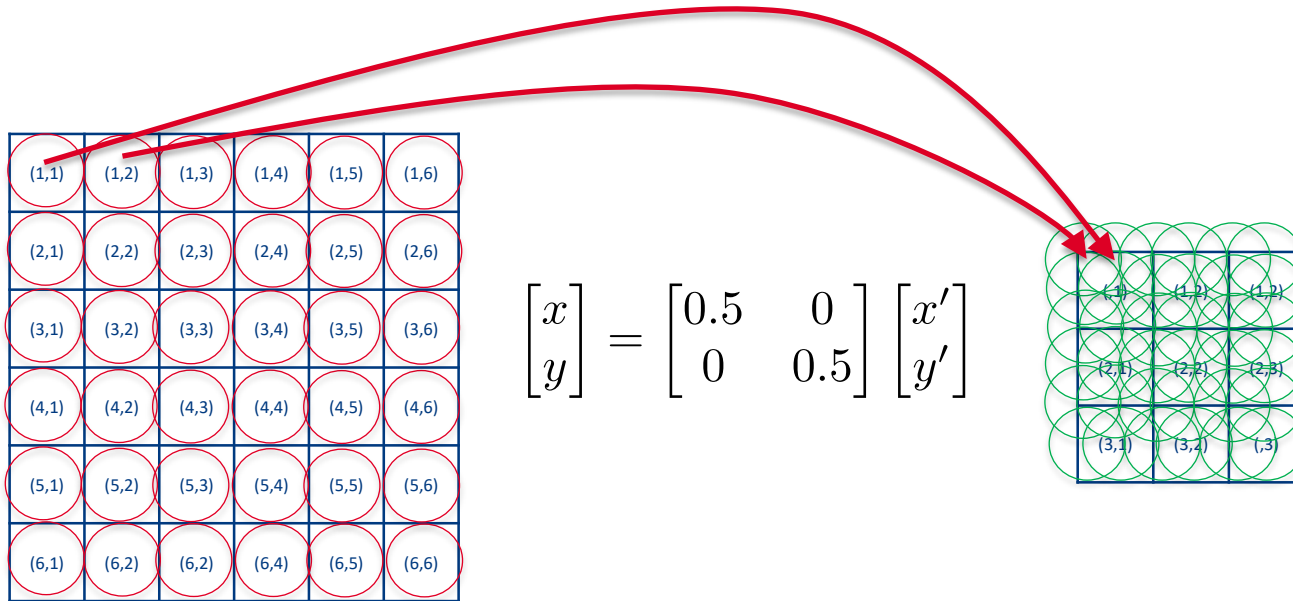
Forward mapping

The problem with forward mapping of the coordinates: **gaps**.



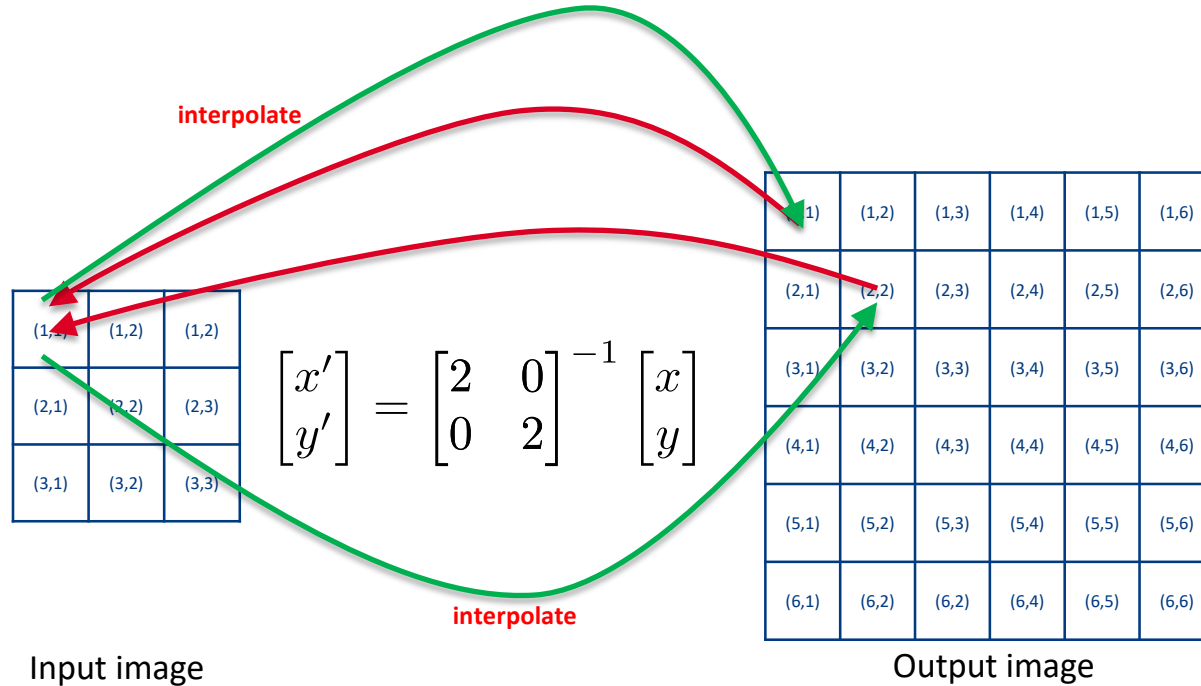
Forward mapping

The problem with forward mapping of the coordinates: **overlaps**.



Inverse mapping

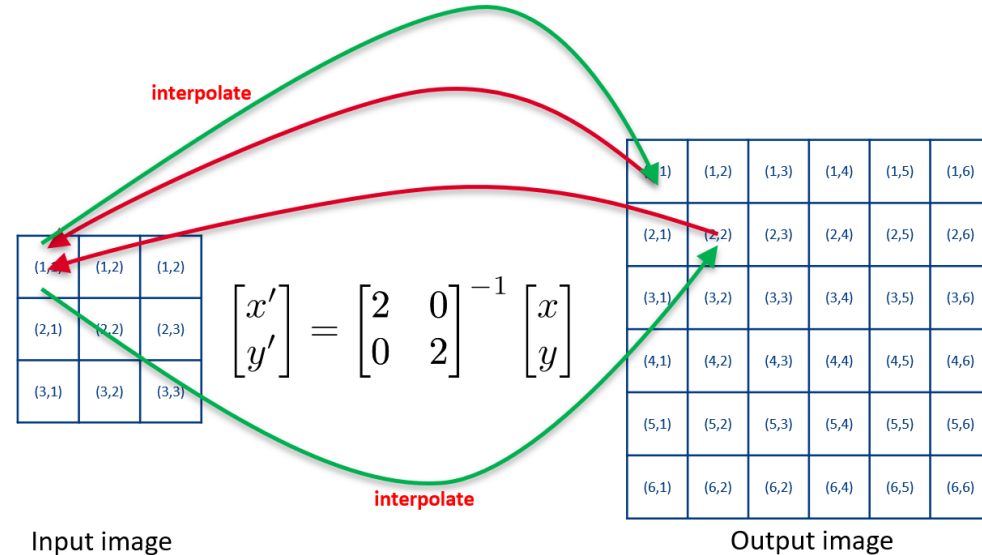
Gaps and multiple values can be avoided by performing **inverse mapping and interpolation**.



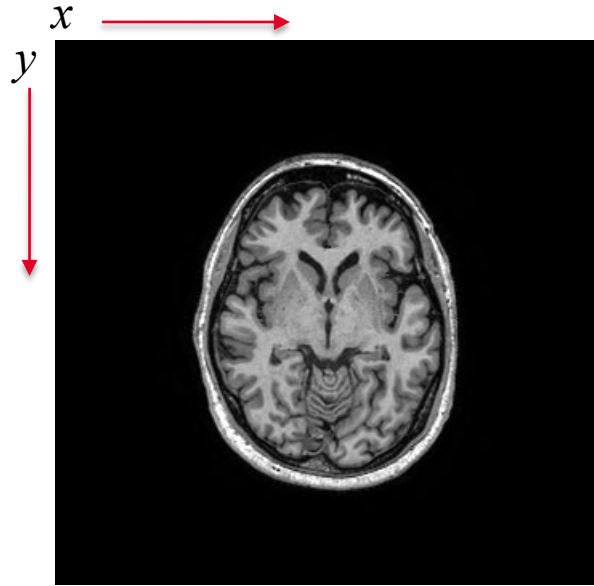
Inverse mapping

Transforming an image with the transformation $\mathbf{T}$ by inverse mapping:

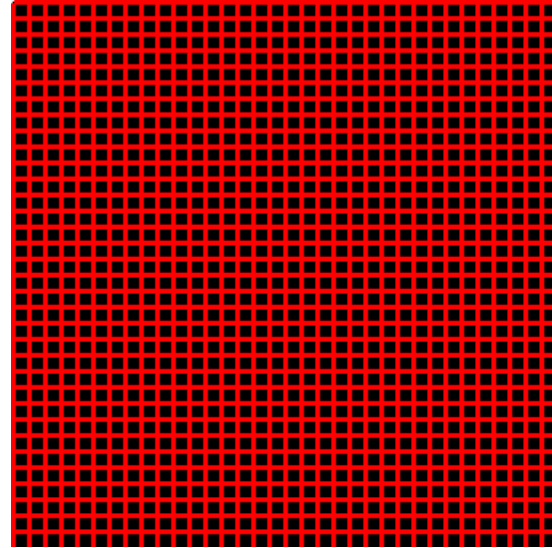
1. Define the grid of the output image
2. Map the points on the grid to the input image with $\mathbf{T}^{-1}$
3. Determine the intensity value at those locations with image interpolation



Example - Inverse mapping

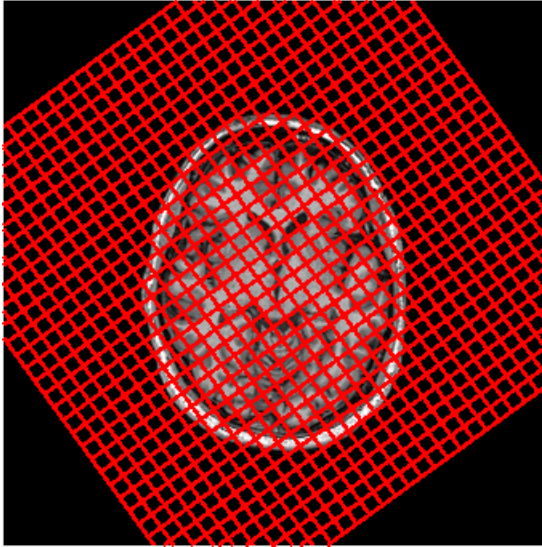


Input image

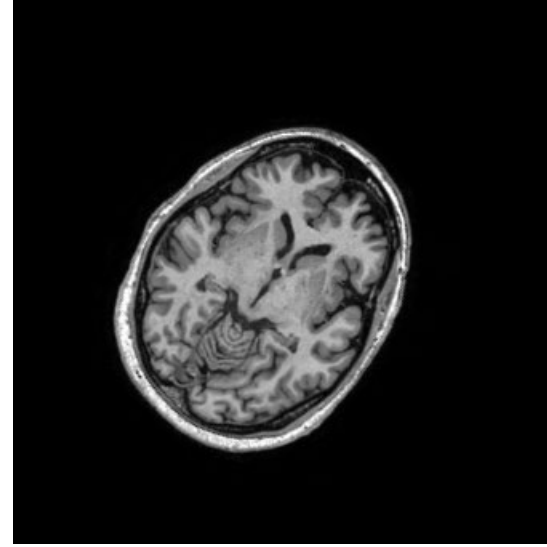


Transformed image (now empty) –
we want a $\pi/5$ rotation of the input

Example - Inverse mapping



Grid of the output transformed
with the inverse transformation

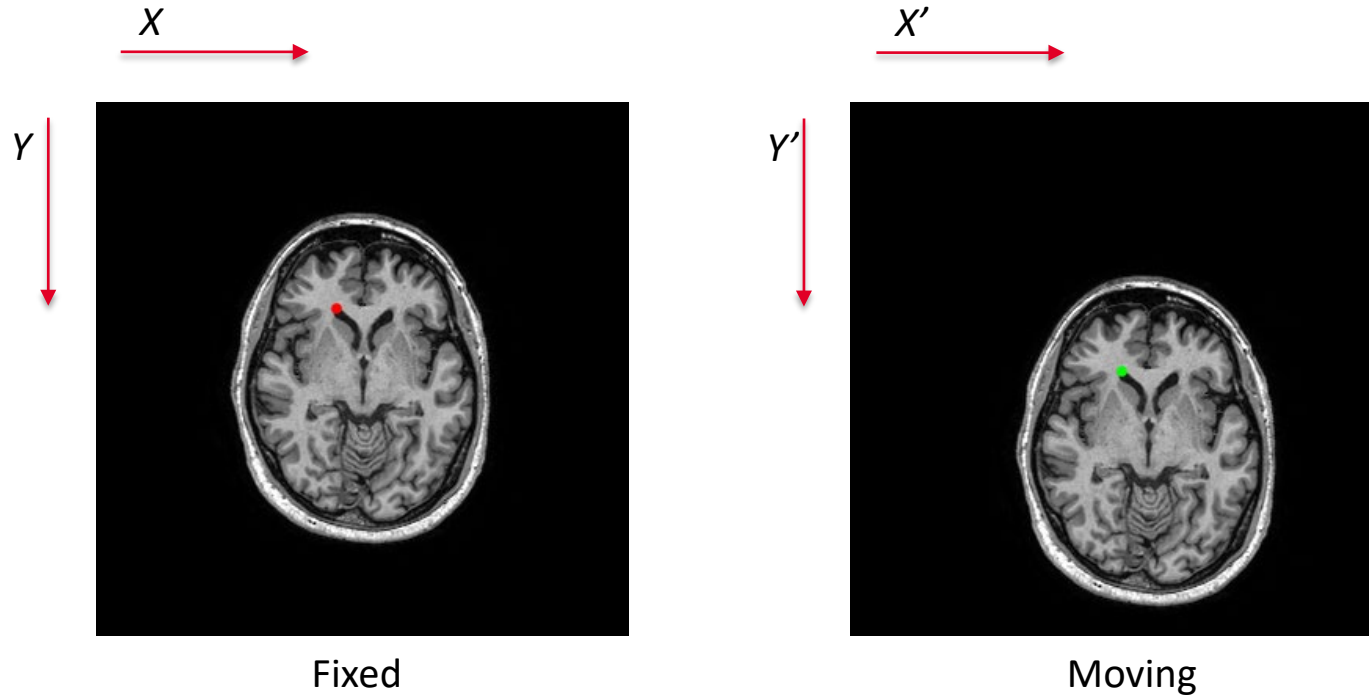


Output image

Outline

- Image transformation
- Point-based registration
- Evaluation metrics for point-based registration

Example – point-based registration



Example – point-based registration

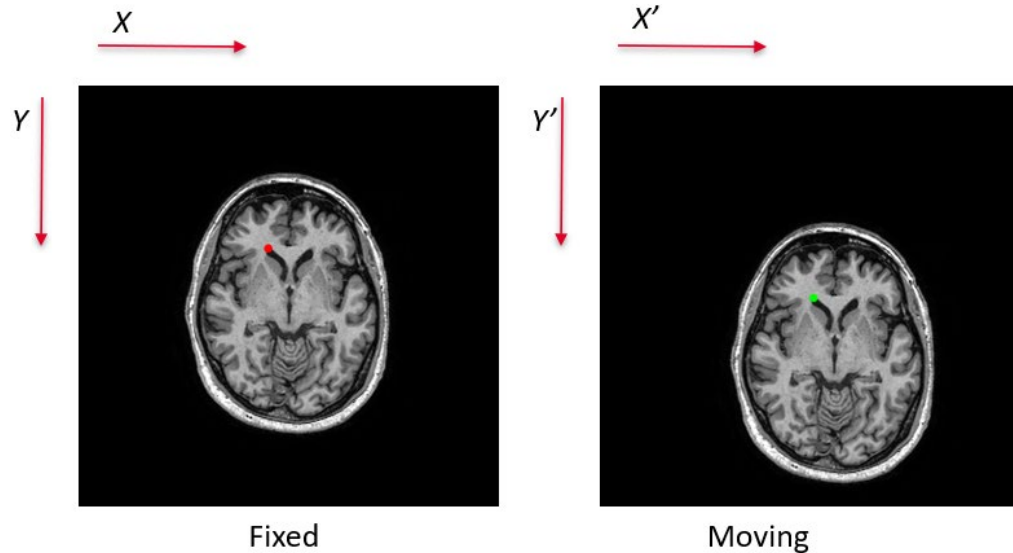
Assumption:

Two images are misaligned only with a **translation**:

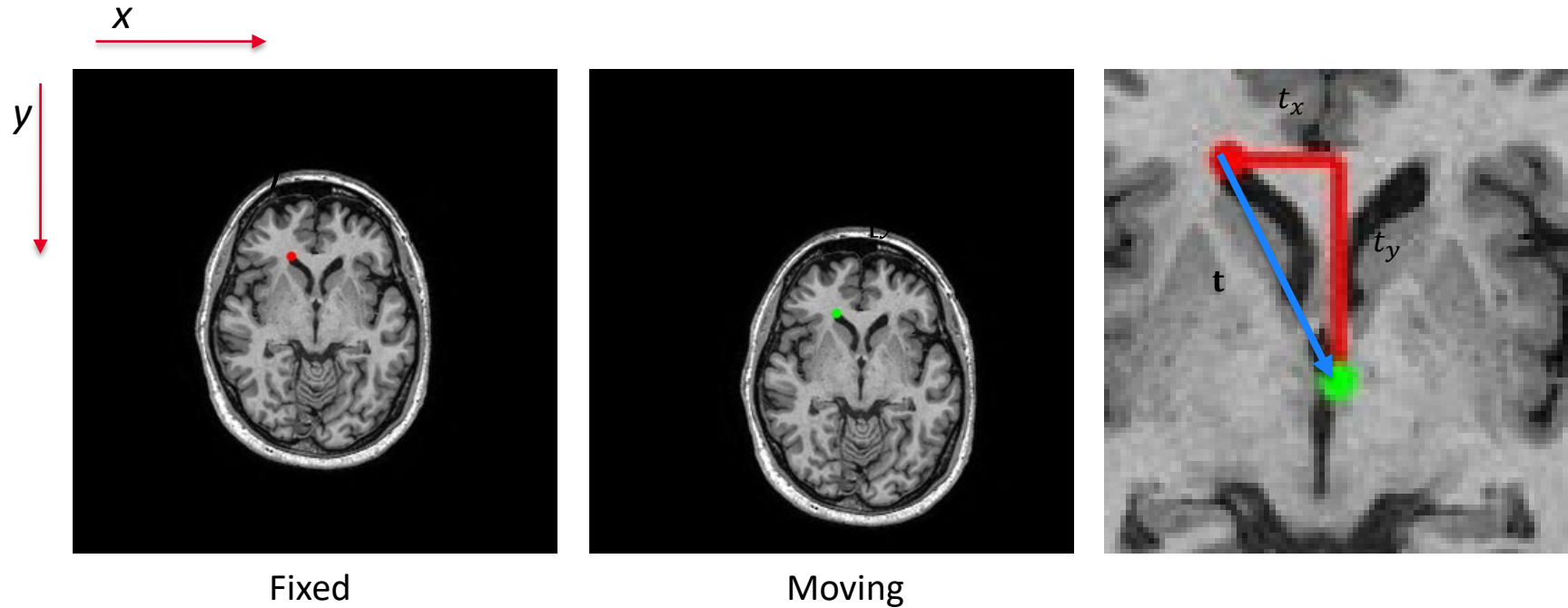
$$\mathbf{x}' = \mathbf{x} + \mathbf{t}$$

Solution:

1. Mark the location of some well **discernable feature** in the fixed image.
2. Mark the corresponding location in the moving image.
3. Compute the translation as: $\mathbf{t} = \mathbf{x}' - \mathbf{x}$
4. Transform the moving image by translating it by $-\mathbf{t}$



Example – point-based registration



Fiducial points

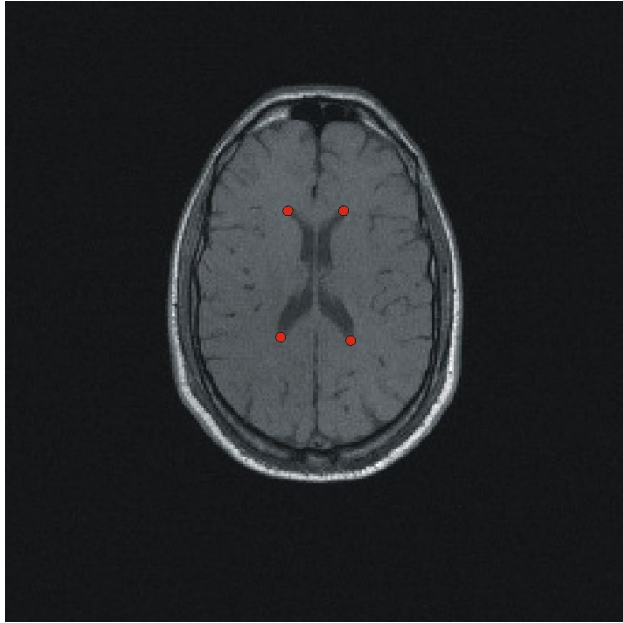
Such points that are taken to be reliable for image registration are called **fiducial points** or **fiducials**.

Fiducials can be placed at either intrinsic or extrinsic features.

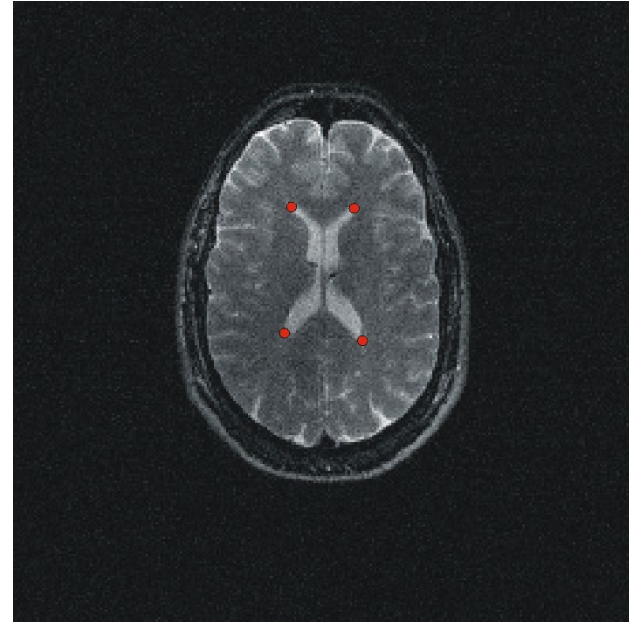
- **Intrinsic features:** e.g. anatomical landmarks.
- **Extrinsic features:** e.g. implanted markers.



Intrinsic fiducials

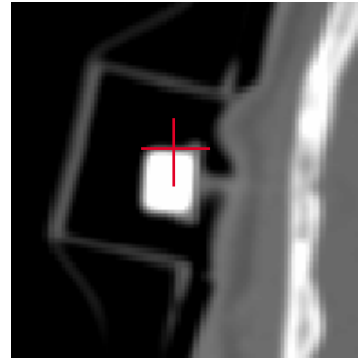
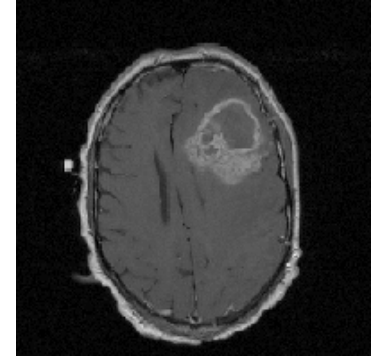
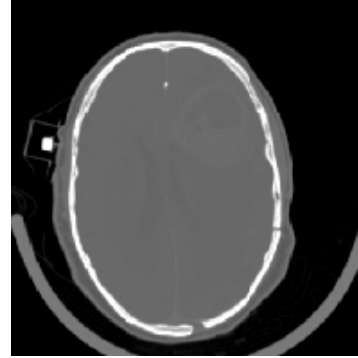
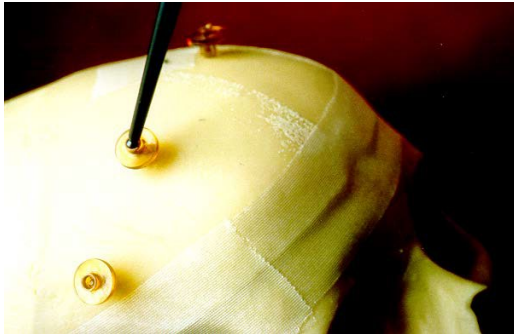


MR T1 weighted

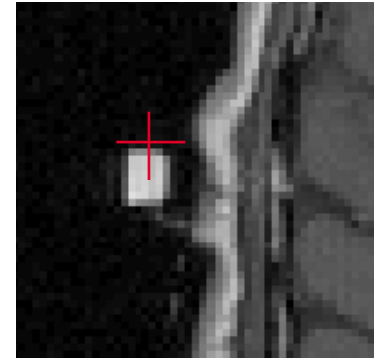


MR T2 weighted

Extrinsic fiducials



CT



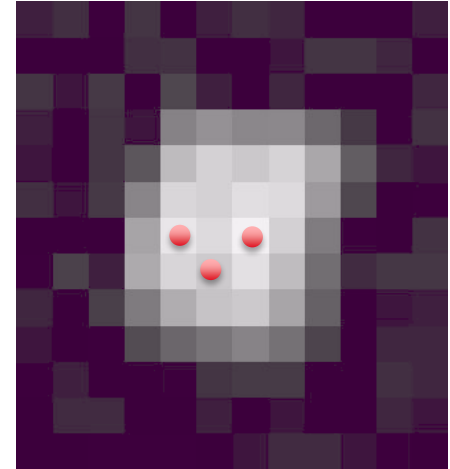
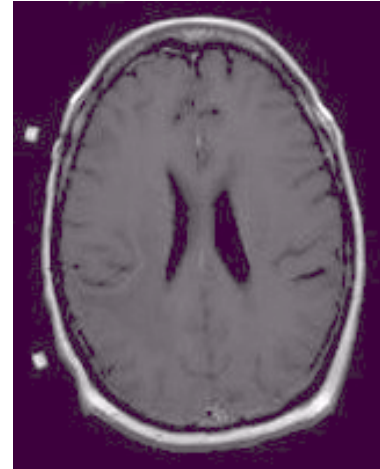
MR

Fiducial alignment

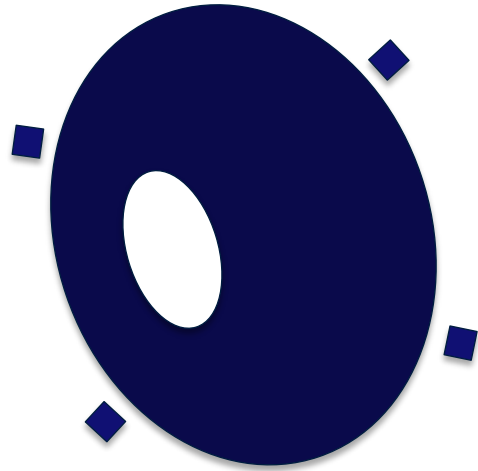
A perfect alignment of the fiducials is usually not possible.

Reasons:

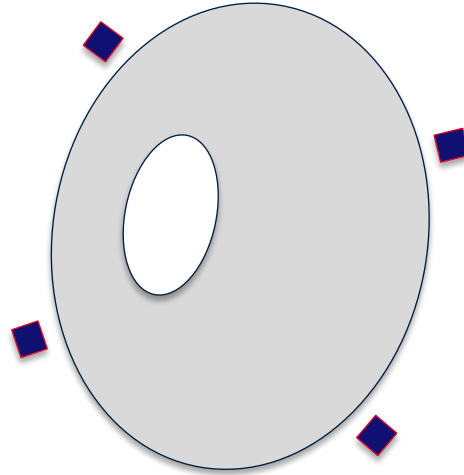
- Localization error of the marker (a.k.a. fiducial localization error, FLE)
- Image distortion
- Shift of extrinsic markers
- Incorrect model assumption
- ...



Fiducial registration error (FRE)

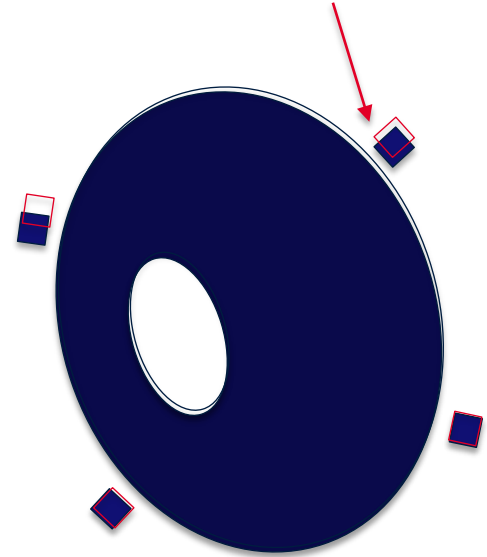


Fixed image



Moving image

Fiducial registration error (FRE)



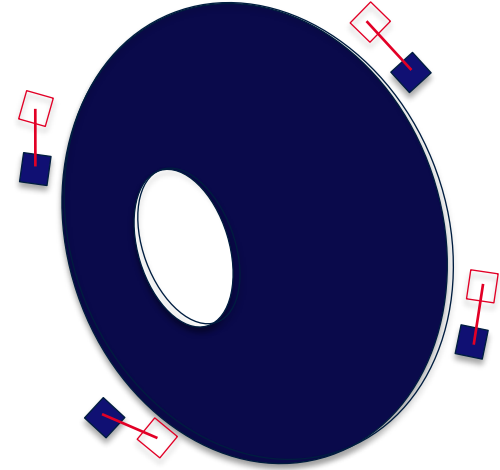
Registered images

Point-based registration - Goal

It is thus not realistic to look for an algorithm that will find a transformation that results in a **perfect alignment** of all corresponding point pairs.

However, we can design an algorithm that will find a transformation that results in the **best possible alignment** given that fact that there will always be some error.

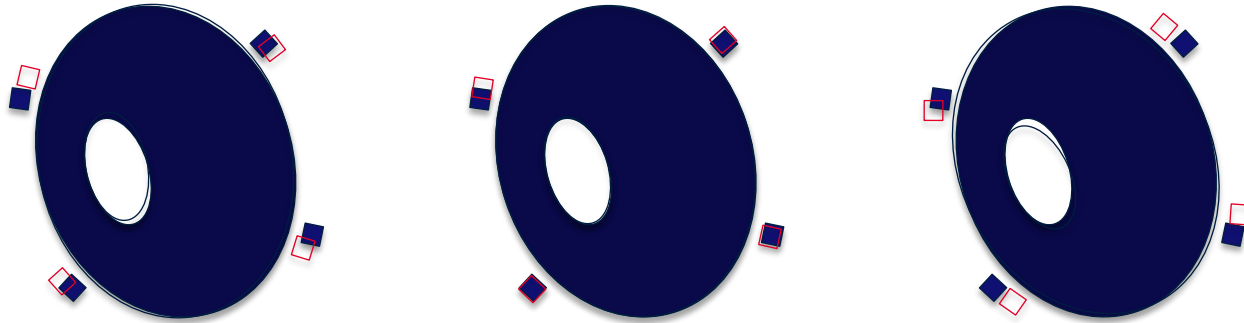
Best possible alignment = lowest error.



Point-based registration – How?

How to find such a transformation?

1. Write the error as a function of the transformation.
2. Find the minimum of the error function w.r.t. the transformation.



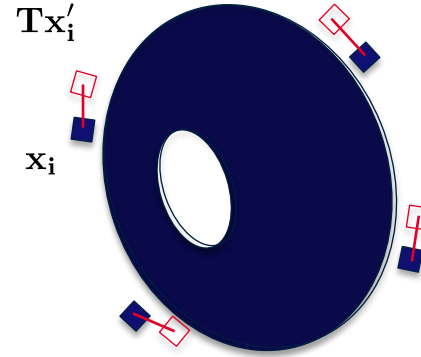
Different transformations will result in different errors.
Our goal: find the transformation that results in the lowest error.

Point-based registration – How?

Step 1: write the error as a function of the transformation.

$$E(\mathbf{T}) = \sum_{i=1}^n \|\mathbf{T}\mathbf{x}'_i - \mathbf{x}_i\|_2^2$$

affine registration $\mathbf{T} = \begin{bmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{bmatrix}$



Point-based registration – How?

Step 2: find the minimum of the error w.r.t. to the parameters.

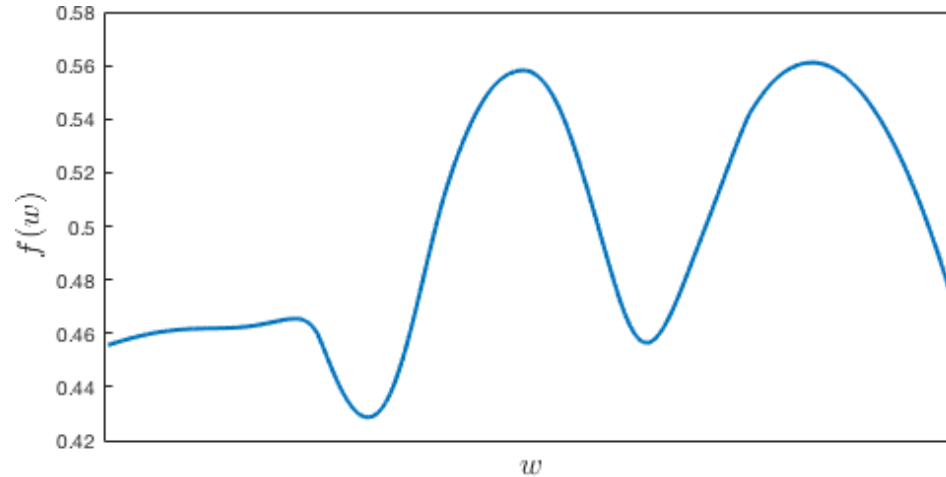
$$\hat{\mathbf{T}} = \arg \min_{\mathbf{T} \in A} \sum_{i=1}^n \|\mathbf{T}\mathbf{x}'_i - \mathbf{x}_i\|_2^2 \quad \mathbf{T} = \begin{bmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

The optimal $\mathbf{T}$ is the one
that minimizes the error

$\mathbf{T}$ belongs to the set of all
affine transformations

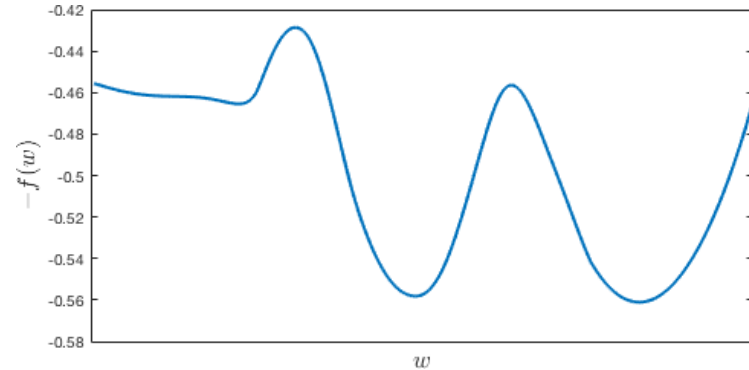
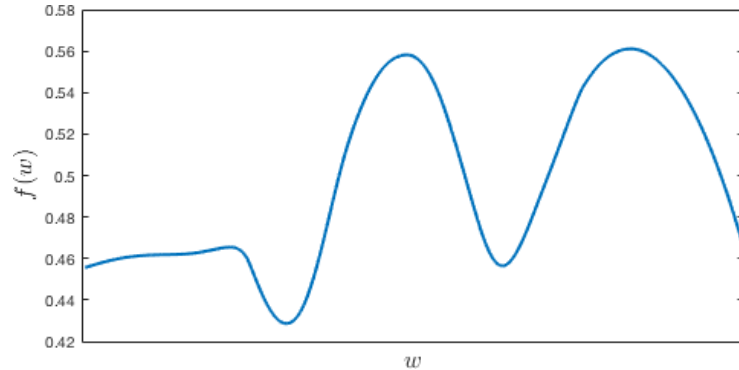
Optimization

Optimization involves finding the “best” parameters according to an “objective function”, which is either minimised or maximised.



Optimization

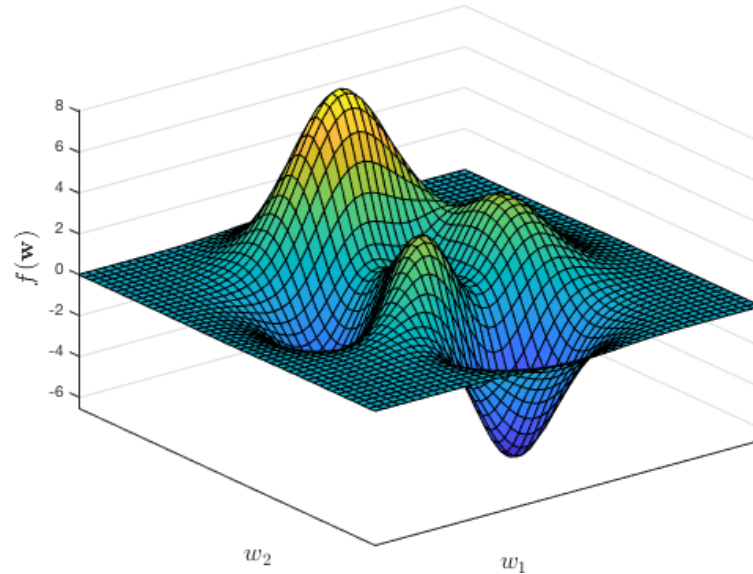
If we have a method that finds the maximum of a function, it can be easily used to find a minimum by inverting the function.



Optimization

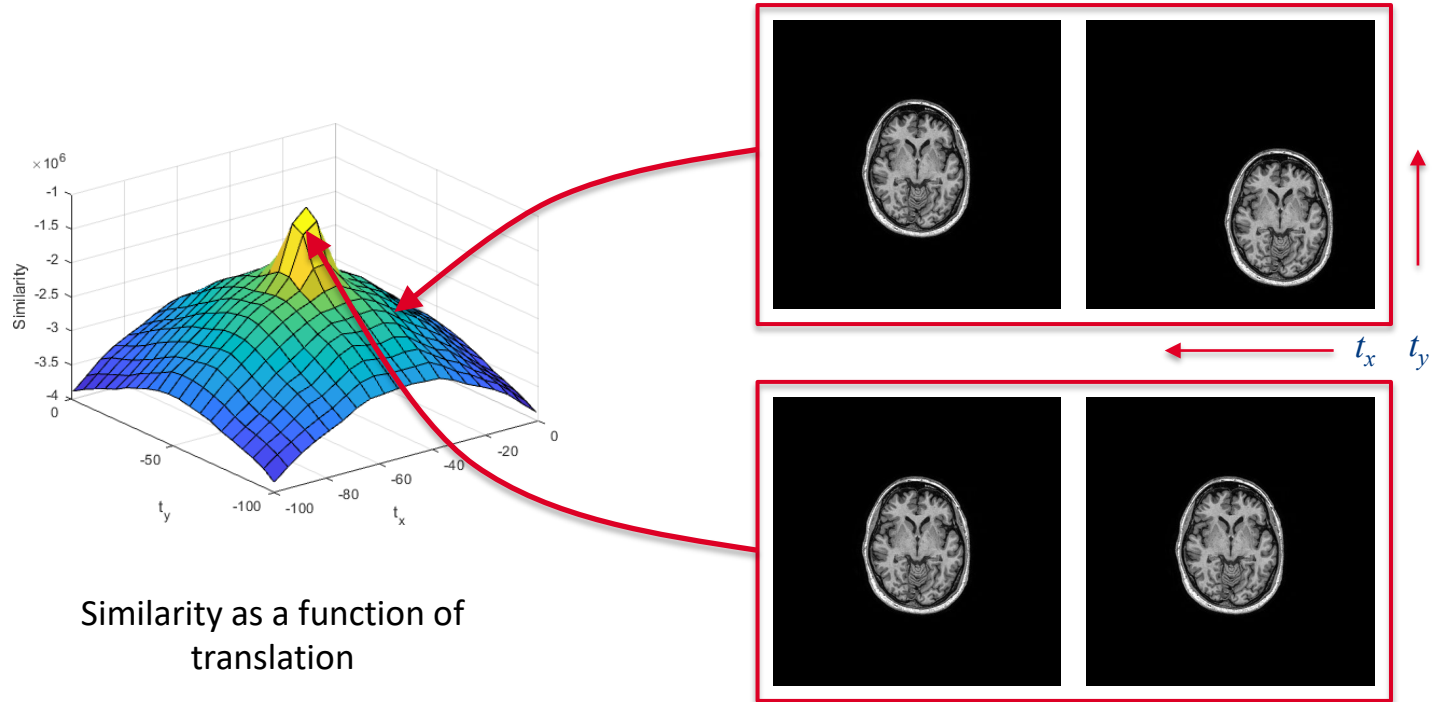
Optimization involves finding some “best” parameters according to an “objective function”, which is either minimised or maximised.

2D example:



Optimization

One approach to optimization is **full search** of the parameter space:



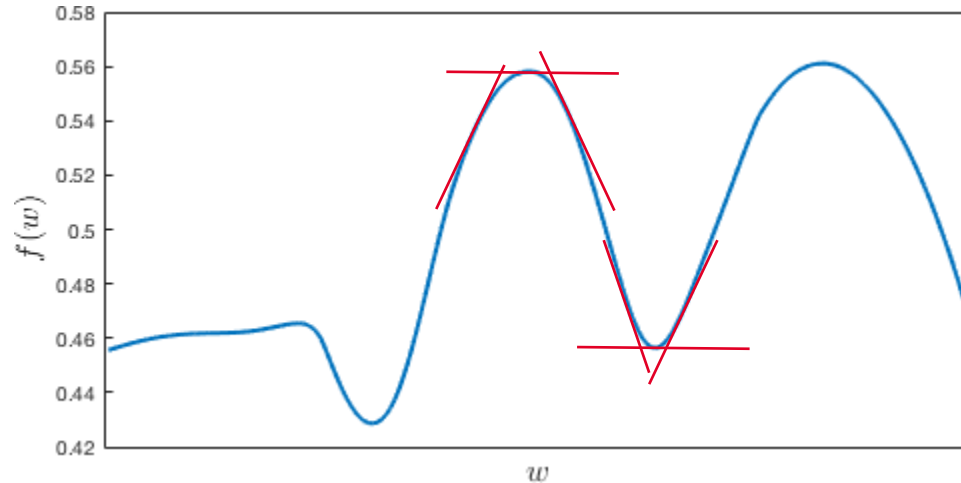
Optimization

However, this is computationally expensive, even for a small number of parameter.

- Consider affine registration of 2D images that has 6 parameters. If we want to evaluate every parameter at 20 values (e.g. 20 rotation angles), all possible combinations would be $20^6 = 64\,000\,000$.
- If one evaluation takes just **0.01 second**, it would take just **over a week** to register a pair of images.

Optimization

How to find the min. and max. of this function **analytically**?

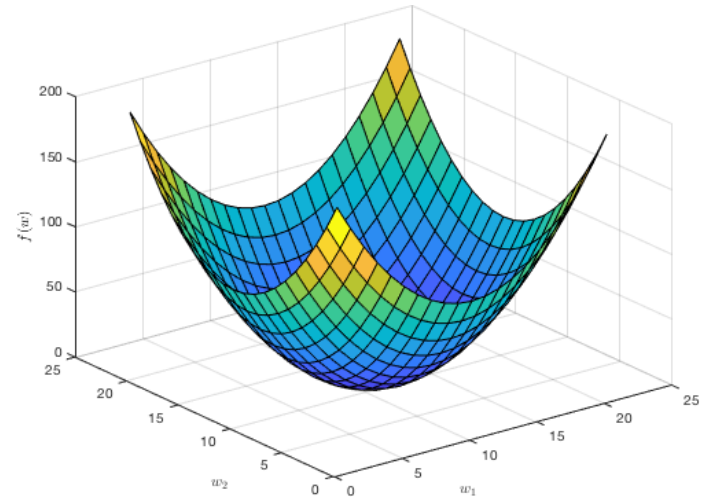
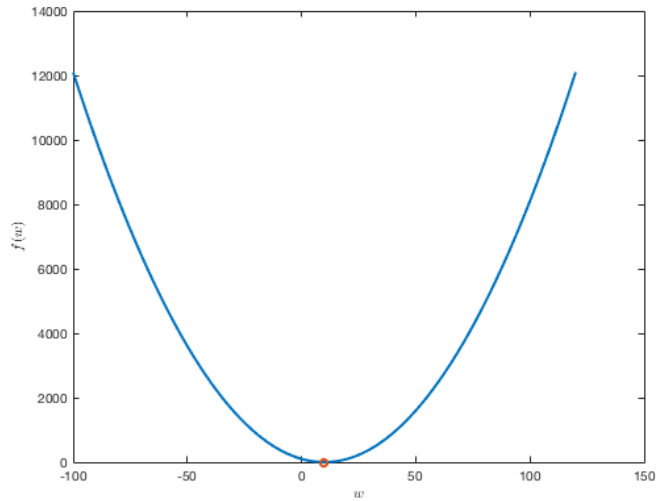


Compute the derivative and set it to zero.

If the function has more than one variable, set the partial derivatives (or gradient vector) to zero.

Optimization

Quadratic functions:



Optimization (affine transformation)

Step 2, find the minimum of the error w.r.t. to the parameters.

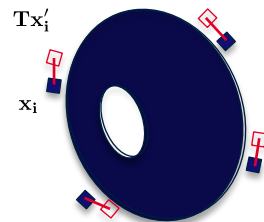
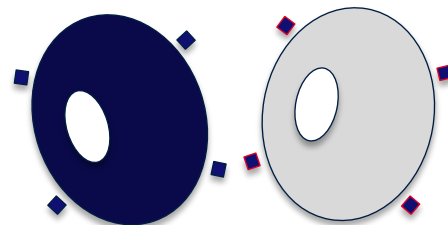
$$E(\mathbf{T}) = \sum_{i=1}^n \|\mathbf{T}\mathbf{x}'_i - \mathbf{x}_i\|_2^2$$

$$E(\mathbf{T}) = \|\mathbf{T}\mathbf{X}' - \mathbf{X}\|_F^2 \quad \leftarrow \text{Frobenius norm}$$

$$\hat{\mathbf{T}} = \arg \min_{\mathbf{T} \in A} \|\mathbf{T}\mathbf{X}' - \mathbf{X}\|_F^2$$

$$\mathbf{X} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \dots \quad \mathbf{x}_i \quad \dots \quad \mathbf{x}_n]$$

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,i} & \dots & x_{1,n} \\ x_{2,1} & x_{2,2} & \dots & x_{2,i} & \dots & x_{2,n} \\ 1 & 1 & \dots & 1 & \dots & 1 \end{bmatrix}$$



Optimization (affine transformation)

Step 2, find the minimum of the error w.r.t. to the parameters.

$$E(\mathbf{T}) = ||\mathbf{T}\mathbf{X}' - \mathbf{X}||_F^2$$

Set of equations:

$$\nabla_{\mathbf{T}} E(\mathbf{T}) = 0$$

Solution for affine transformation:

$$\mathbf{T} = \mathbf{X}\mathbf{X}'^T(\mathbf{X}'\mathbf{X}'^T)^{-1}$$

$$\mathbf{X} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \dots \quad \mathbf{x}_i \quad \dots \quad \mathbf{x}_n]$$

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,i} & \dots & x_{1,n} \\ x_{2,1} & x_{2,2} & \dots & x_{2,i} & \dots & x_{2,n} \\ 1 & 1 & \dots & 1 & \dots & 1 \end{bmatrix}$$

Optimization (rigid transformation)

Can we use the same approach to **find the parameters of a rigid registration** (rotation and translation only)?

- **Step 1**, write the error as a function of the transformation:

$$E(\mathbf{T}) = \sum_{i=1}^n \|\mathbf{T}\mathbf{x}'_i - \mathbf{x}_i\|_2^2$$

- **Step 2**, find the minimum of the error function w.r.t. the transformation:

$$\hat{\mathbf{T}} = \arg \min_{\mathbf{T} \in R} \sum_{i=1}^n \|\mathbf{T}\mathbf{x}'_i - \mathbf{x}_i\|_2^2$$

Optimization (rigid transformation)

Compared with affine registration, the solution is now constrained to transformation matrices that only do **rotation** and **translation**:

$$\mathbf{T} = \begin{bmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{T} = \begin{bmatrix} \cos(\phi) & -\sin(\phi) & t_x \\ \sin(\phi) & \cos(\phi) & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

Affine

Rigid

Using the same approach as before will **NOT** guarantee rigid transformation.

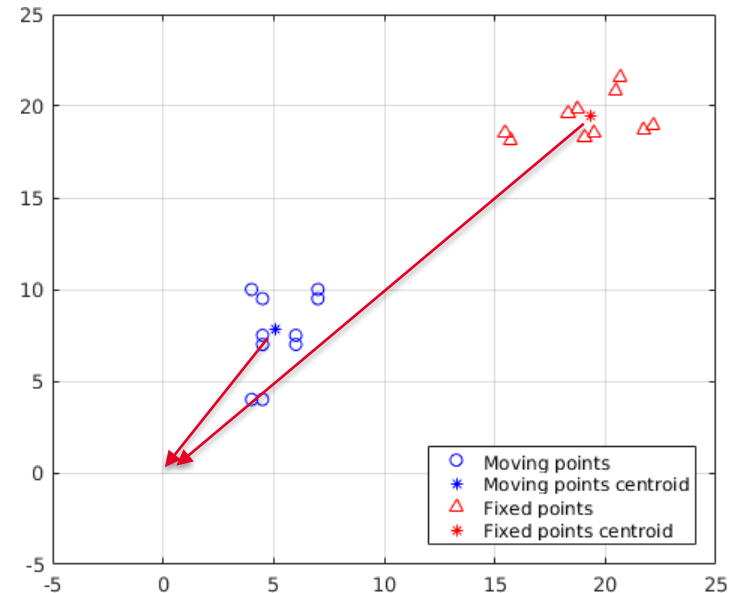
Optimization (rigid transformation)

The minimization of the error of rigid registration is called “**orthogonal Procrustes problem**”.

Algorithm:

1. Compute the centroids of the fiducial points in the fixed and moving image.

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i; \quad \bar{\mathbf{x}}' = \frac{1}{n} \sum_{i=1}^n \mathbf{x}'_i$$

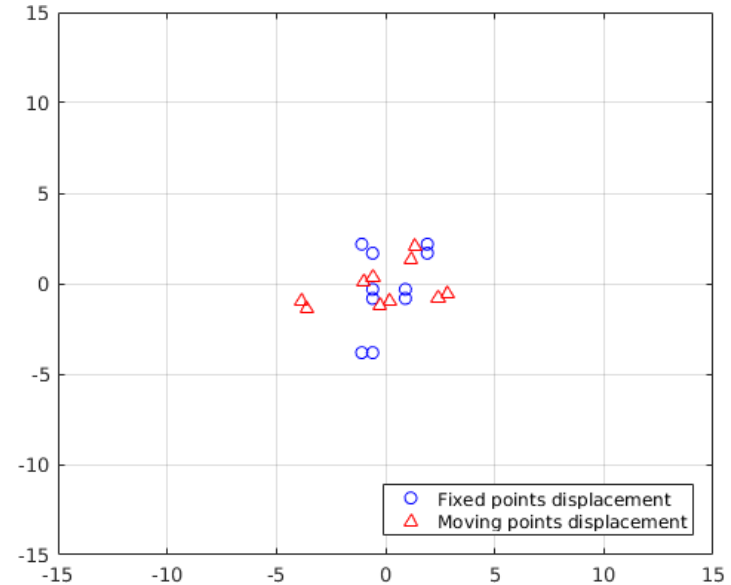


Optimization (rigid transformation)

2. Compute the displacements:

$$\hat{\mathbf{x}}_i = \mathbf{x}_i - \bar{\mathbf{x}}$$

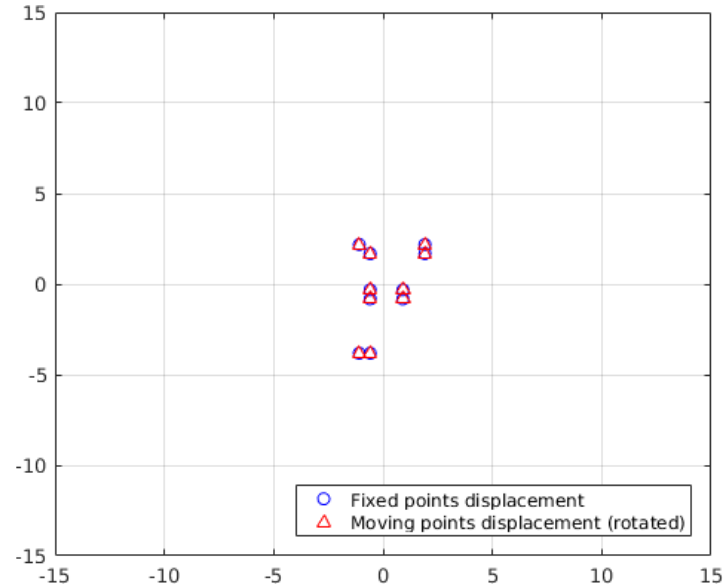
$$\hat{\mathbf{x}}'_i = \mathbf{x}'_i - \bar{\mathbf{x}}'$$



Optimization (rigid transformation)

3. Compute the rotation matrix:

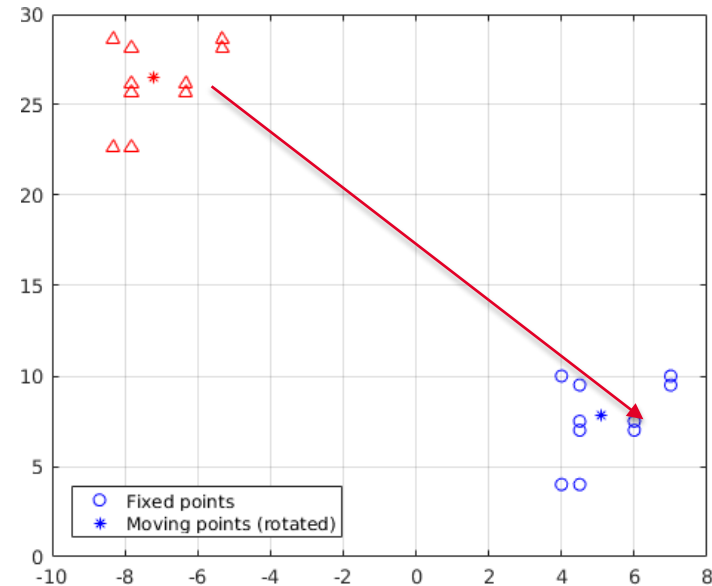
$$\mathbf{R} = \mathbf{V} \text{diag}(1, 1, \mathbf{V}\mathbf{U}) \mathbf{U}^\top$$



Optimization (rigid transformation)

4. Compute the translation:

$$\mathbf{t} = \bar{\mathbf{x}} - \mathbf{R}\bar{\mathbf{x}}'$$



Optimization (rigid transformation)

1. Compute the centroids of the fiducial points in the fixed and moving image.

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i; \quad \bar{\mathbf{x}}' = \frac{1}{n} \sum_{i=1}^n \mathbf{x}'_i$$

2. Compute the displacement of each fiducial point

$$\hat{\mathbf{x}}_i = \mathbf{x}_i - \bar{\mathbf{x}}; \quad \hat{\mathbf{x}}'_i = \mathbf{x}'_i - \bar{\mathbf{x}}'$$

3. Compute the covariance matrix of the fiducials

$$\mathbf{H} = \sum_{i=1}^n \hat{\mathbf{x}}'_i \hat{\mathbf{x}}_i^\top$$

4. Perform singular value decomposition of the covariance matrix.

$$\mathbf{H} = \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top$$

5. Compute the rotation matrix:

$$\mathbf{R} = \mathbf{V} \text{diag}(1, 1, \mathbf{VU}) \mathbf{U}^\top$$

→ The diagonal term ensures that we get a proper rotation matrix with determinant equal 1.

6. Compute the translation vector:

$$\mathbf{t} = \bar{\mathbf{x}} - \mathbf{R} \bar{\mathbf{x}}'$$

Optimization (unknown correspondence)

Thus far we assumed that the correspondence between the points in the moving and fixed images is known. What if we do not know the correspondence?

We introduce a **correspondence function** that is now **subject to optimization**:

$$E(\mathbf{T}) = \sum_{i=1}^n \|\mathbf{TC}(\mathbf{x}_i, \{\mathbf{x}'_j\}) - \mathbf{x}_i\|^2$$

Iterative optimization

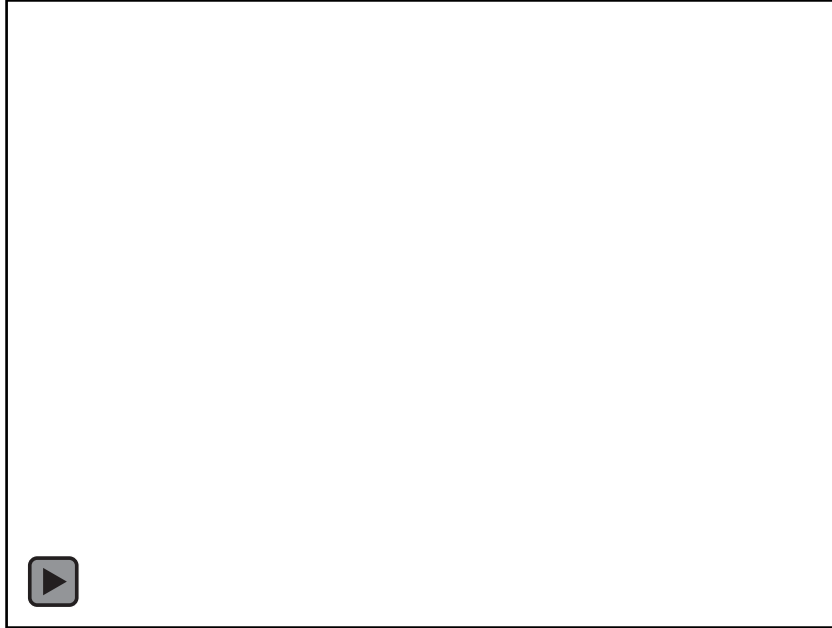
This leads to optimization problems that are usually solved with an **iterative procedure**.

Example: **iterative closest point**.

Iterate until convergence:

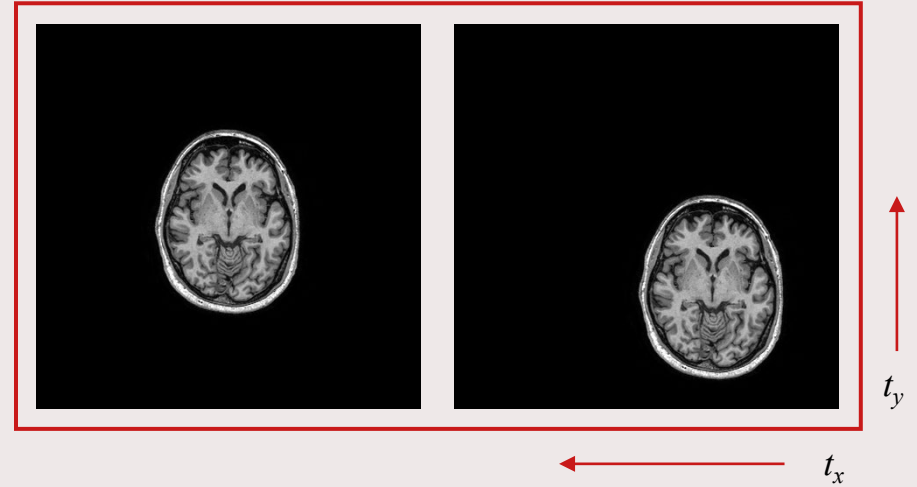
- For each point in the moving shape find the closest point in the fixed shape.
- Assume the found correspondence is correct and register (using one of the algorithms that work with correspondence).

Iterative closest point

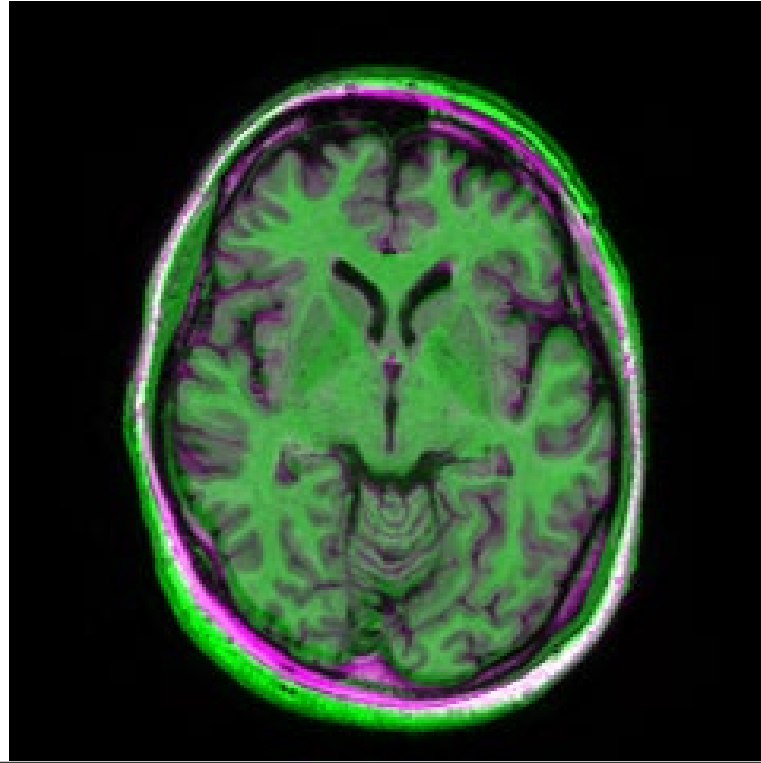
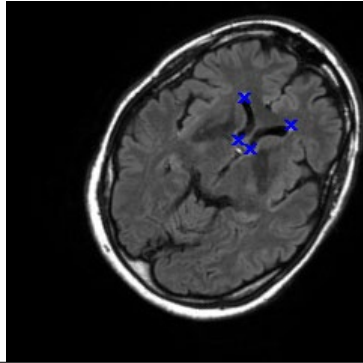
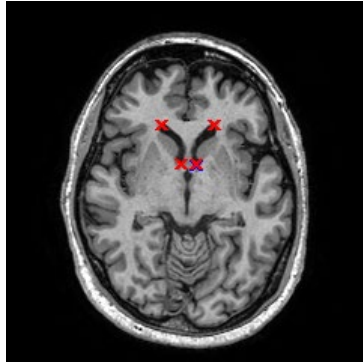


Outline

- Image transformation
- Point-based image registration
- Evaluation of registration accuracy



Evaluation of registration accuracy



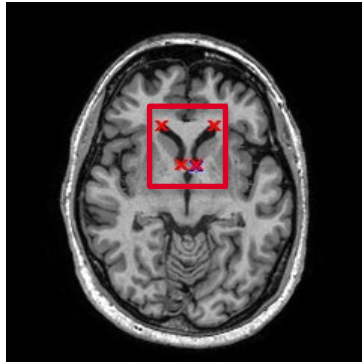
Evaluation of registration accuracy

Image registration can be evaluated by computing the registration error for some target corresponding point pairs.

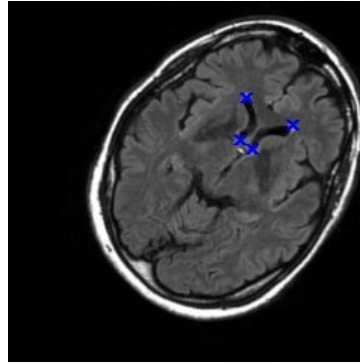
→ Target registration error (TRE)

- The target points should be selected in locations that are relevant for some treatment or diagnosis.
- Basically, this is the same as the procedure for selecting corresponding point pairs to compute the transformation.
- However, the target corresponding points must be different from the points used to compute the transformation! Why?

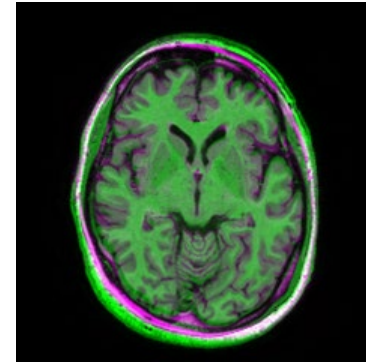
(Poor) Example of affine registration:



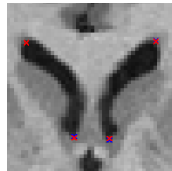
Fixed



Moving



Registered



Note that the registration error for the fiducials is very low. This is because we minimize this error! The error outside of these locations is larger.

Evaluation procedure

1. Perform image registration (compute the transformation matrix T)
2. Annotate some target corresponding point pairs in the fixed and moving images. These must be different from the corresponding points used to compute T and at locations that are relevant for some treatment or diagnosis.
3. Transform the points from the moving image
4. Compute the target registration error as the average distance between the points in the fixed image and the transformed moving points.

Summary

The student can:

- explain the role of inverse mapping and interpolation when transforming an image and compute inverse transformation matrices
- design a general algorithm to register two images based on fiducials (i.e., composing an error function and minimizing this function w.r.t. the transformation T)
- use optimization to find the minimum of this error function
- recall the exact solution for T (matrix notation) when constrained to affine registration
- give at least four reasons why perfect alignment of multiple fiducials is not possible in practice
- describe the algorithm required to find the optimal parameters of T when constrained to rigid registration (orthogonal Procrustes problem)
- explain the principle behind the iterative closest point algorithm
- use the target registration error (TRE) to evaluate image registration



Feedback

Thank you!

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Next: intensity-based registration